

Program : Diploma in Information Technology	
Course Code : 4266	Course Title: Network Switching Technologies Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide practical exposure about Network Switching Technologies from a Network Administrator's point-of-view.
- To provide basic concept of Local Area Network, Network Switching and its Administration.

Course Prerequisites:

Topic / Description	Course Code	Course Title	Semester
Basic knowledge in Network Simulator		Computer Hardware and Networking Lab	3
Subnetting, LAN, IP Address, Switching etc		Fundamentals of Network Administration	3

Course Outcomes

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Configure a network switch using basic Internetwork Operating System commands.	10	Applying
CO2	Setup a Virtual Local Area Network (VLAN)	11	Applying
CO3	Configure access and trunk port, VLAN Trunking Protocol (VTP) and Port Security.	10	Applying
CO4	Configure Access-Control List (ACL) and Network Address Translation (NAT).	11	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Configure a network switch using basic Internetwork Operating System commands.		
M1.01	Make use of various <i>show</i> commands in a switch	2	Applying
M1.02	Make use of <i>copy</i> , <i>write</i> , <i>erase</i> and <i>delete</i> command	2	Applying
M1.03	Experiment with configuration of IP addresses in a switch	2	Applying
M1.04	Experiment with <i>Telnet</i> , <i>SSH</i> and <i>DHCP</i> services in a Switch	2	Applying
M1.05	Utilize <i>Backup</i> and <i>Restoration</i> of configuration in a Switch.	2	Applying
CO2	Setup a Virtual Local Area Network (VLAN)		
M2.01	Experiment with MAC Address Table verification	2	Applying
M2.02	Experiment with creating and renaming VLANs in a Switch	3	Applying
M2.03	Experiment with assigning VLAN membership to <i>switchports</i>	2	Applying
M2.04	Experiment with verifying VLAN information by <i>show vlan brief</i> command	2	Applying
M2.05	Experiment with testing the reachability between the users of VLANs	2	Applying
	Lab Exam – I	1.5	

CO3	Configure access and trunk port, VLAN Trunking Protocol (VTP) and Port Security.		
M3.01	Experiment with configuring <i>access</i> and <i>trunk</i> port in switches	4	Applying
M3.02	Experiment with configuring VTP	3	Applying
M3.03	Experiment with configuring Port Security in a Switch	3	Applying
CO4	Configure Access-Control List (ACL) and Network Address Translation (NAT).		
M4.01	Make use of creating and defining permit/deny statements in a standard ACL using <i>access-list</i> command	3	Applying
M4.02	Experiment with configuring Static and Dynamic NAT	4	Applying
M4.03	Open-ended experiment. **	4	Applying
	Lab Exam – II	1.5	

**** Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Construct campus network using Network Simulator, then make VLAN for each department.
2. Construct a hospital network using Network Simulator, then do inter VLAN routing.

Text / Reference:

T/R	Book Title / Author
T1	Todd Lammle, <i>CCNA Routing and Switching Complete Study Guide</i> , 2 nd Edition, Sybex (Wiley) Publication, 2016
R1	Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. <i>Elements of Computer Networking: An Integrated Approach</i> . CareerMonk Publications, 2014.

Online Resources:

Sl. No.	Website Link
1	https://www.freeccnastudyguide.com/study-guides/icnd1-100-101/
2	https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-tutorials/
3	https://www.guru99.com/introduction-ccna.html
4	https://study-ccna.com/
5	http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php